

Mobile Sales Performance & Customer Analytics Dashboard

Power BI Business Intelligence & Data Analytics Project Report

1. Executive Summary

The **Mobile Sales Performance & Customer Analytics Dashboard** is a Power BI-based business intelligence project developed to analyze mobile sales transactions and understand sales performance, product contribution, geographic performance, customer purchasing behavior, payment preferences, and time-based sales patterns.

The project uses a dataset containing **3,835 transaction records covering the period from 2021 to 2024**, across **5 mobile brands, 15 mobile models, 19 cities, and 4 payment methods**. The dataset also includes customer-related attributes, customer ratings, transaction information, units sold, pricing information, and date-related fields.

The project follows an end-to-end data analytics workflow. The raw dataset was first inspected and validated for data quality issues such as duplicate records, missing values, inconsistent categorical values, incorrect data types, and inconsistencies in date-related fields. The cleaned data was then transformed using **Power Query** and prepared for analysis in Power BI.

Several **DAX measures** were developed to calculate key performance indicators such as Total Sales, Total Quantity, Transactions, and Average Price. These metrics were then incorporated into an interactive dashboard containing KPI cards, trend analysis, geographic analysis, brand and model comparisons, payment method analysis, customer rating analysis, and day-of-week analysis.

The analysis identified several notable patterns. **Apple generated the highest overall sales value among the five brands**, while **Delhi emerged as the strongest-performing city**. **UPI recorded the highest transaction volume among the four payment methods**. These findings demonstrate how transactional data can be transformed into meaningful business intelligence to support performance monitoring and data-driven decision-making.

2. Introduction

2.1 Background

The mobile phone industry generates large volumes of transactional data through customer purchases, product sales, pricing, payment methods, locations, and customer feedback.

While raw transaction data contains valuable information, it is difficult for business stakeholders to interpret large datasets directly. Business intelligence tools such as Power BI can help transform this information into interactive dashboards that provide a clear understanding of business performance.

This project focuses on analyzing mobile sales data to identify patterns across different business dimensions, including:

- Brand
- Mobile model
- City
- Payment method
- Customer ratings
- Sales quantity
- Price
- Transaction activity
- Time period
- Day of week

The objective is not only to visualize the data but also to use analytical techniques to answer practical business questions and generate actionable insights.

3. Project Objectives

The main objectives of the project are:

1. To clean and validate raw mobile sales transaction data.
 2. To prepare the dataset for business intelligence analysis.
 3. To develop meaningful sales KPIs using DAX.
 4. To analyze sales performance across different mobile brands.
 5. To identify high-performing mobile models.
 6. To analyze geographic sales performance across cities.
 7. To understand customer payment preferences.
 8. To analyze customer rating patterns.
 9. To identify time-based and day-of-week sales patterns.
 10. To develop an interactive Power BI dashboard.
 11. To convert raw transactional data into meaningful business insights.
 12. To provide a foundation for data-driven business decision-making.
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4. Business Problem

Businesses selling mobile devices need to understand not only how much they sell but also **where, what, when, and how customers are purchasing products**.

Without a consolidated analytical view, it can be difficult to answer important questions such as:

- Which mobile brands generate the highest sales?
- Which products contribute most to revenue?
- Which cities are the strongest markets?
- What payment methods do customers prefer?
- How many units are being sold?
- What is the average selling price?
- How does sales activity change over time?
- Which days show higher sales activity?
- What does the customer rating distribution look like?

The project addresses these challenges by developing an interactive Power BI dashboard that brings multiple dimensions of sales and customer data together in a single analytical environment.

5. Dataset Description

The dataset contains **3,835 mobile sales transactions** covering the period **2021–2024**.

Dataset Coverage

Attribute	Details
Total Transactions	3,835

Attribute	Details
Time Period	2021–2024
Mobile Brands	5
Mobile Models	15
Cities	19
Payment Methods	4
Customer Information	Available
Customer Ratings	Available
Transaction Information	Available
Sales Information	Available

Major Data Categories

The dataset contains fields related to:

- Transaction identification
- Date
- Brand
- Mobile model
- City
- Customer information
- Payment method
- Units sold
- Price per unit
- Customer rating
- Day name
- Sales transaction information

6. Data Preparation & Data Cleaning

Data quality is an important part of any analytics project. Before developing the Power BI dashboard, the source data was inspected and prepared to ensure that the resulting analysis was reliable.

6.1 Data Quality Checks

The following checks were performed:

Dataset Structure

The dataset structure was reviewed to understand the available fields, their data types, and their relevance to the analysis.

Transaction ID Validation

Transaction IDs were reviewed to ensure that transactions could be uniquely identified and that duplicate transaction records could be detected.

Duplicate Check

The dataset was checked for duplicate records to prevent duplicate transactions from affecting sales calculations and dashboard KPIs.

Missing Value Check

Columns were reviewed for blank, null, or missing values. Where applicable, missing values were evaluated to determine whether they could affect the analysis.

Data Type Validation

Numerical and date-related fields were reviewed to ensure that they were stored using appropriate data types.

Examples include:

- Units Sold → Numeric
- Price Per Unit → Numeric
- Date → Date
- Customer Rating → Numeric

Categorical Standardization

Categorical values were reviewed and standardized to avoid inconsistencies caused by differences in spelling, formatting, or representation.

Day Name Validation

Date-related fields and Day Name values were reviewed to ensure consistency for day-of-week analysis.

7. Data Transformation

After data validation, the dataset was prepared for Power BI analysis.

Power Query Transformation

Power Query was used to support the data preparation process, including:

- Cleaning source data
- Standardizing data types
- Transforming relevant fields
- Preparing categorical fields
- Validating analytical columns
- Preparing data for visualization
- Creating a business-ready dataset

The goal of this stage was to ensure that the dataset was structured appropriately before developing the analytical measures and dashboard.

8. Analytical Approach

The project follows an end-to-end analytical workflow:

Raw Data → Data Inspection → Data Cleaning → Data Transformation → DAX Measures → Visualization → Business Analysis → Insights

This approach ensures that the dashboard is based on structured and validated data rather than directly visualizing unprocessed source information.

9. DAX Measures

DAX was used to create the primary business metrics required for dashboard analysis.

9.1 Total Sales

Total Sales =

```
SUMX(  
    Sales_Data,  
    Sales_Data[Units Sold] * Sales_Data[Price Per Unit]  
)
```

Purpose

The Total Sales measure calculates the overall sales value by multiplying units sold by price per unit for each transaction.

9.2 Total Quantity

Total Quantity =

```
SUM(Sales_Data[Units Sold])
```

Purpose

This measure calculates the total number of mobile units sold across all transactions.

9.3 Transactions

Transactions =

```
COUNTROWS(Sales_Data)
```

Purpose

This measure calculates the total number of transaction records included in the analysis.

9.4 Average Price

Average =

```
AVERAGE(Sales_Data[Price Per Unit])
```

Purpose

This measure calculates the average selling price per mobile unit across the dataset.

10. Dashboard Design

The Power BI dashboard was designed to provide a high-level overview while allowing users to explore specific dimensions of sales and customer behavior.

10.1 KPI Cards

The dashboard includes KPI cards displaying:

- **Total Sales**
- **Total Quantity**
- **Transactions**

- **Average Price**

These KPIs provide stakeholders with an immediate understanding of overall business performance.



Figure 1: Key Performance Indicators

11. Sales Trend Analysis

The sales trend analysis provides a time-based view of sales activity across the dataset.

This analysis helps stakeholders understand:

- Changes in sales activity
- High and low sales periods
- Overall sales movement
- Potential patterns across the analyzed period

The time dimension covers **2021–2024**, enabling users to explore sales activity over multiple years.



Figure 2: Sales Trend Analysis

12. Geographic Sales Analysis

Geographic analysis was performed across **19 cities** to identify differences in sales performance by location.

This analysis helps answer:

- Which cities generate the highest sales?
- Where is sales activity concentrated?
- Which locations represent stronger markets?

Key Finding

Delhi emerged as the strongest-performing city, with approximately **₹203.9M** in sales.

Mumbai followed with approximately **₹127.2M**.

Business Interpretation

The difference between the leading cities and other locations indicates that sales are not evenly distributed geographically. This suggests that geographic segmentation could be important for future sales planning, marketing allocation, and market expansion analysis.

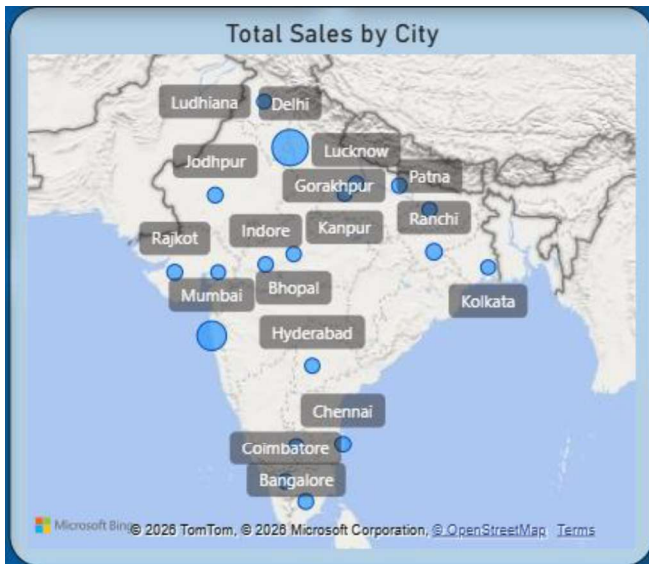


Figure 3: Geographic Sales Performance

13. Brand Performance Analysis

The dashboard compares the performance of the five mobile brands.

Brand Sales Performance

Brand **Approx. Sales**

Apple ₹161.6M

Samsung ₹160.0M

OnePlus ₹153.7M

Vivo ₹150.1M

Xiaomi ₹143.8M

Key Finding

Apple recorded the highest overall sales value among the five analyzed brands, followed by Samsung.

Xiaomi recorded the lowest sales value among the five brands.

Business Interpretation

The relatively close sales performance across the brands indicates a competitive distribution of sales rather than an extreme concentration around a single brand.

The analysis also demonstrates the importance of comparing both **sales value and quantity**, as a brand generating high revenue may not necessarily have the highest unit volume.



Figure 4: Brand Performance Analysis

14. Mobile Model Analysis

The dashboard provides a detailed comparison of mobile models to identify products contributing most significantly to sales.

The analysis allows users to evaluate:

- High-performing models
- Sales contribution by model
- Unit volume
- Relative product performance

Business Importance

Model-level analysis can help businesses identify products that may require:

- Greater inventory availability
- Additional promotional activity
- Pricing evaluation
- Performance monitoring

15. Payment Method Analysis

The dashboard analyzes customer transactions across four payment methods:

- UPI
- Debit Card
- Credit Card
- Cash

Transaction Distribution

Payment Method Approx. Transactions

UPI	1,011
Debit Card	948
Credit Card	947
Cash	929

Key Finding

UPI recorded the highest number of transactions, with approximately 1,011 transactions.

Business Interpretation

The higher transaction volume through UPI indicates strong adoption of digital payment methods within the analyzed dataset.

This information can be useful for businesses when evaluating payment infrastructure, customer convenience, and digital payment strategies.

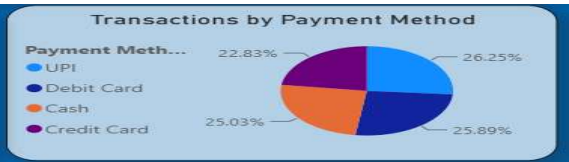


Figure 6: Payment Method Analysis

16. Customer Rating Analysis

Customer ratings were analyzed to understand the distribution of customer feedback across transactions.

The rating analysis provides a view of:

- Rating distribution
- Common rating levels
- Overall customer feedback patterns

Customer ratings can be used as an additional dimension when evaluating product performance and customer experience.



Figure 7: Customer Rating Distribution

17. Day-of-Week Analysis

The dashboard analyzes sales activity by day of the week.

This helps identify:

- Higher-activity days
- Lower-activity days
- Differences in transaction activity across weekdays
- Potential purchasing patterns

Day-level analysis can provide useful information for promotional planning, staffing, and sales scheduling.



Figure 8: Day-of-Week Sales Analysis

18. Interactive Dashboard Features

The dashboard includes interactive filters that allow users to dynamically explore the dataset.

Users can filter the analysis by:

- Brand
- Mobile Model

- Payment Method
- Day Name

These filters allow stakeholders to drill down from overall performance into specific product, payment, and time segments.



Example

A user can select a specific brand and mobile model and then analyze:

- Sales
- Quantity
- Transactions
- Average price
- Customer ratings
- Geographic distribution

This interactive functionality makes the dashboard suitable for exploratory business analysis.

19. Business Questions & Answers

The dashboard was designed around practical business questions.

Sales Performance

What is the overall sales performance?

The dashboard provides a consolidated view of total sales, quantity sold, transaction volume, and average selling price.

How many mobile units were sold?

The Total Quantity KPI provides the total number of mobile units sold across the dataset.

How many transactions occurred?

The Transactions KPI provides the total number of transaction records analyzed.

What is the average selling price?

The Average Price KPI provides the average price per mobile unit.

Product Performance

Which mobile brand generates the highest sales?

Apple generated the highest overall sales value at approximately ₹161.6M.

Which mobile models perform best?

The mobile model analysis identifies the products contributing most to overall sales and quantity.

Which brands generate the highest sales quantity?

Brand-level quantity analysis enables comparison of sales volume across the five mobile brands.

Which products contribute most to overall revenue?

Model-level analysis highlights the products that contribute most significantly to overall sales value.